



Project Title: Smart City Traffic Control System using IoT and Network Simulation

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1. Abstract

This paper illustrates the planning and simulation of a Smart City Traffic Control Network with the help of Cisco Packet Tracer. The brand-new network layout is designed as several star-based LANs, where each city district is equipped with a dedicated access switch that serves as the central point of connection. These LANs are bound together by three routers, and communication between the routers is carried out through two serial WAN links that use /30 subnets.

The network is capable of handling different types of usage requirements, such as office workstations, publicly available wireless access, printing services, technical support resources, and voice endpoints. Moreover, a dedicated segment is employed to illustrate the connection to the outside world through a DSL modem. The addressing plan gives each district a separate /24 subnet, which facilitates better organization and makes troubleshooting simpler, while routing ensures that all districts can communicate with each other. To cap it all, the network operation is confirmed via ping tests, wireless connectivity checks, and VoIP verification.

2. Introduction

Stable and scalable networks are the backbone of modern smart city environments that facilitate the seamless interconnection of departments, public areas, and service zones. Although star topology is the typical practice at the LAN level due to its simplicity of management, it is frequently resorted to: end devices are connected to a single switch, faults can be located more quickly, and expansion is easier.

So, the city network is the city network is implemented as a group of star networks in a project. Each area (zone) works as a separate LAN with a central switch, while routers are used to provide connectivity across zones. The design serves the demonstration of voice communication through wired users (PCs and laptops), wireless users (public Wi-Fi areas), shared resources (printers and a server), and IP phones. A report describes the topology, the IP plan, the configuration strategy, and the testing procedures for checking correctness.



3. Network Summary and Goals

3.1 Network Summary

The network is divided into several functional zones, each placed into its own subnet. Switches provide local connectivity inside each zone. Routers interconnect the zones and forward traffic between different networks. Wireless access points are used where mobility is required, and VoIP endpoints are added to represent voice communication needs.

3.2 Project Goals

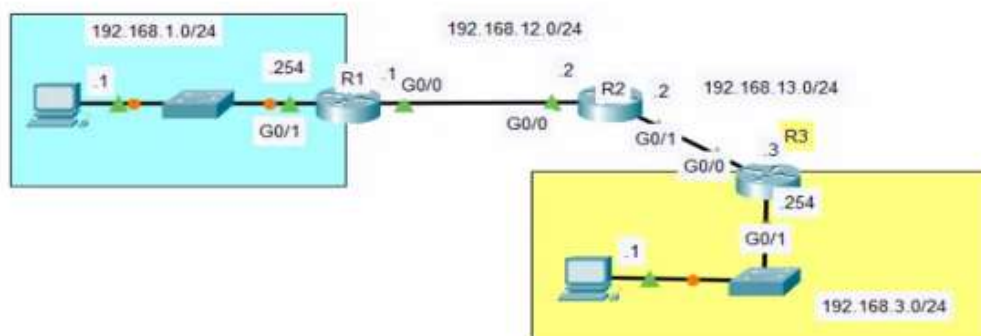
1. Build a multi-zone network where each zone is a separate star-topology LAN.
2. Apply a clear IP addressing scheme using distinct /24 networks per zone.
3. Configure routing so every subnet can reach all other subnets.
4. Deploy wireless access in specific zones and confirm wireless clients can join and communicate.
5. Include VoIP endpoints and validate reachability and calls.
6. Simulate external connectivity using a DSL modem segment.
7. Verify end-to-end connectivity using ping and Packet Tracer simulation tools.

4. Chosen Network Architecture

The adopted architecture is a **hierarchical star design**:

- At the access layer: each zone is a star LAN centered on a switch.
- At the distribution/core: routers connect these LANs and provide inter-subnet routing.
- Between routers: serial point-to-point WAN links are used with /30 addressing to reduce IP waste and maintain clean point-to-point routing.

This approach improves clarity and scalability: adding a new zone typically requires adding a new switch + subnet and connecting it to a router interface.



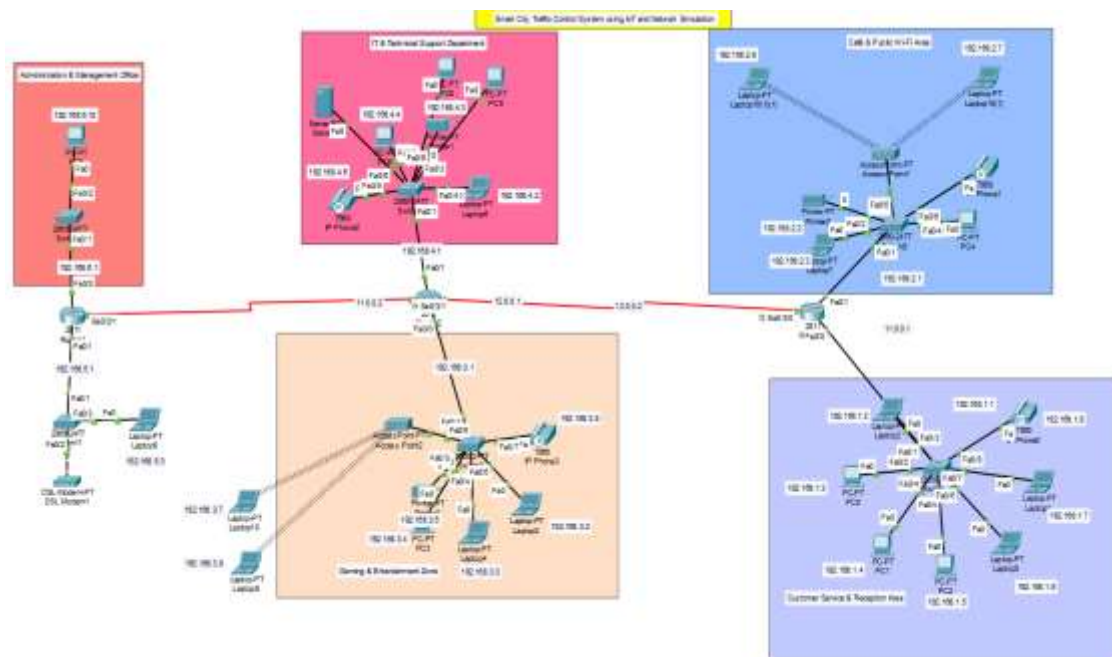


The network is organized into the following segments (each one a star LAN):

- **Customer Service & Reception Area – 192.168.1.0/24**
Workstations and laptops for customer-facing services + IP phone.
Gateway: 192.168.1.1
- **Café & Public Wi-Fi Area – 192.168.2.0/24**
A public wireless zone via AP, with a local printer, PC, and IP phone.
Gateway: 192.168.2.1
- **Gaming & Entertainment Zone – 192.168.3.0/24**
Mixed wired and wireless devices, a printer, and IP phone.
Gateway: 192.168.3.1
- **IT & Technical Support Department – 192.168.4.0/24**
Technical PCs, a server for services/resources, a printer, and IP phone.
Gateway: 192.168.4.1
- **Server/External Connectivity Segment – 192.168.5.0/24**
Represents connectivity testing toward an outside network using DSL modem.
Gateway: 192.168.5.1
- **Administration & Management Office – 192.168.6.0/24**
Management workstation for administrative tasks.
Gateway: 192.168.6.1

WAN Links

- Router1 ↔ Router2 : 11.0.0.0/30
- Router2 ↔ Router3 : 12.0.0.0/30





5. Network Devices and Components

Table 1: Device Inventory (Summary)

Device Type	Count
Routers	3
Switches	6
Servers	1
DSL Modems	1
PCs	7
Laptops	8
IP Phones	4
Printers	3
Wireless Access Points	2

6. IP Plan and Subnet Allocation

Table 2: Router Interfaces

Device	Interface	IP Address	Subnet Mask
Router1	Fa0/0 (Admin LAN)	192.168.6.1	255.255.255.0
Router1	Fa0/1 (External/DSL LAN)	192.168.5.1	255.255.255.0
Router1	Se0/3/1 (to Router2)	11.0.0.1	255.255.255.252
Router2	Fa0/1 (IT LAN)	192.168.4.1	255.255.255.0
Router2	Fa0/0 (Gaming LAN)	192.168.3.1	255.255.255.0
Router2	Se0/3/1 (to Router1)	11.0.0.2	255.255.255.252
Router2	Se0/3/0 (to Router3)	12.0.0.1	255.255.255.252
Router3	Fa0/0 (Customer Service LAN)	192.168.1.1	255.255.255.0
Router3	Fa0/1 (Public Wi-Fi LAN)	192.168.2.1	255.255.255.0
Router3	Se0/3/0 (to Router2)	12.0.0.2	255.255.255.252

Table 3: WAN Links (/30)

Link	Network	Side	IP	Mask
R1–R2	11.0.0.0/30	Router1	11.0.0.1	255.255.255.252
		Router2	11.0.0.2	255.255.255.252
R2–R3	12.0.0.0/30	Router2	12.0.0.1	255.255.255.252
		Router3	12.0.0.2	255.255.255.252

Table 4: Customer Service & Reception Area (192.168.1.0/24 – GW 192.168.1.1)

Device	IP Address	Mask	GW
Laptop2	192.168.1.2	255.255.255.0	192.168.1.1
PC0	192.168.1.3	255.255.255.0	192.168.1.1
PC1	192.168.1.4	255.255.255.0	192.168.1.1
PC2	192.168.1.5	255.255.255.0	192.168.1.1
Laptop0	192.168.1.6	255.255.255.0	192.168.1.1
Laptop1	192.168.1.7	255.255.255.0	192.168.1.1



IP Phone0	192.168.1.8	255.255.255.0	192.168.1.1
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Table 5: Cafe & Public Wi-Fi Area (192.168.2.0/24 – GW 192.168.2.1)

Device	IP Address	Mask	GW
Printer2	192.168.2.2	255.255.255.0	192.168.2.1
Laptop7	192.168.2.3	255.255.255.0	192.168.2.1
PC4	192.168.2.4	255.255.255.0	192.168.2.1
IP Phone1	192.168.2.5	255.255.255.0	192.168.2.1
Laptop10 (Wi-Fi)	192.168.2.7	255.255.255.0	192.168.2.1
Laptop10(1) (Wi-Fi)	192.168.2.9	255.255.255.0	192.168.2.1
AccessPoint1	192.168.2.10	255.255.255.0	192.168.2.1

Table 6: Gaming & Entertainment Zone (192.168.3.0/24 – GW 192.168.3.1)

Device	IP Address	Mask	GW
Laptop3	192.168.3.2	255.255.255.0	192.168.3.1
Laptop4	192.168.3.3	255.255.255.0	192.168.3.1
PC3	192.168.3.4	255.255.255.0	192.168.3.1
Printer0	192.168.3.5	255.255.255.0	192.168.3.1
AccessPoint2	192.168.3.6	255.255.255.0	192.168.3.1
Laptop0 (Wi-Fi)	192.168.3.7	255.255.255.0	192.168.3.1
IP Phone3	192.168.3.8	255.255.255.0	192.168.3.1

Table 7: IT & Technical Support Department (192.168.4.0/24 – GW 192.168.4.1)

Device	IP Address	Mask	GW
Laptop8	192.168.4.2	255.255.255.0	192.168.4.1
PC8	192.168.4.3	255.255.255.0	192.168.4.1
PC9	192.168.4.4	255.255.255.0	192.168.4.1
Printer1	192.168.4.5	255.255.255.0	192.168.4.1
Server0	192.168.4.6	255.255.255.0	192.168.4.1
IP Phone2	192.168.4.7	255.255.255.0	192.168.4.1

Table 8: Server/External Connectivity Segment (192.168.5.0/24 – GW 192.168.5.1)

Device	IP Address	Mask	GW
Laptop8	192.168.5.3	255.255.255.0	192.168.5.1
Laptop0	192.168.5.7	255.255.255.0	192.168.5.1
Laptop9	192.168.5.8	255.255.255.0	192.168.5.1

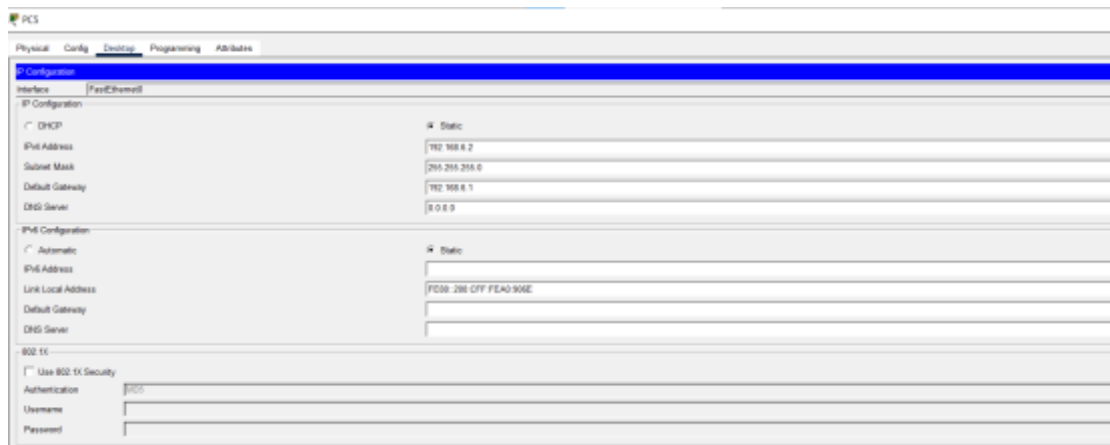


7. Implementation and Configuration Steps

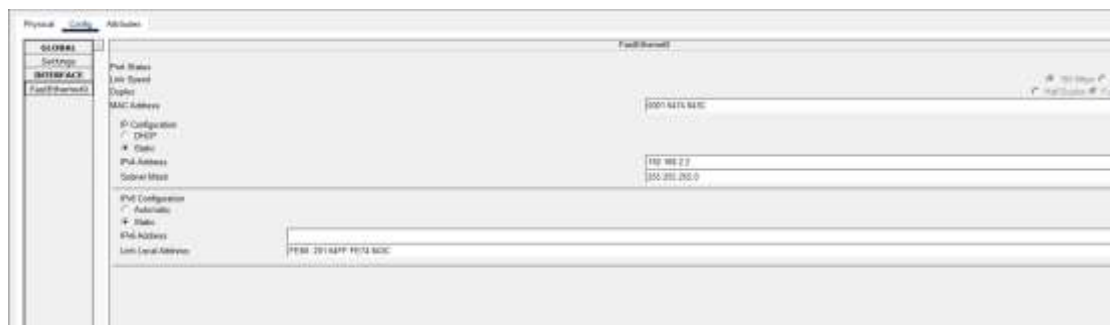
7.1 End-Device Setup

All hosts (PCs, laptops, printers, and IP phones) were configured with the correct addressing details. In areas using Wi-Fi, clients were also connected to the correct SSID.

- **PC/Laptop:** Desktop → IP Configuration



- **Printer:** Desktop → IP Configuration





- **Wireless Client:** Desktop → PC Wireless → select SSID → connect



- **IP Phone:** Config → Network





7.2 Router Setup and Routing Configuration

Routers were configured on LAN-facing interfaces (FastEthernet) and WAN-facing interfaces (Serial). Static routes were used to reach remote LANs via the next hop router across the serial links.

Verification was done using:

- show ip interface brief

```
Router3>
Router3>ena
Router3#
Router3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.1.1	YES	manual	up	up
FastEthernet0/1	192.168.2.1	YES	manual	up	up
Serial0/3/0	11.0.0.1	YES	manual	up	up
Serial0/3/1	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	up	down

```
Router3#
```

- show ip route

```
Router3#sho
Router3#show ip rou
Router3#show ip route
```

Codes: L - local, C - connected, S - static, B - BGP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, Ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```

11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    11.0.0.0/8 is directly connected, Serial0/3/0
L    11.0.0.1/32 is directly connected, Serial0/3/0
L    12.0.0.0/24 is subnetted, 1 subnets
S    12.0.0.0/24 [1/0] via 11.0.0.2
S    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/0
L    192.168.1.1/32 is directly connected, FastEthernet0/0
L    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, FastEthernet0/1
L    192.168.2.1/32 is directly connected, FastEthernet0/1
S    192.168.3.0/24 [1/0] via 11.0.0.2
S    192.168.4.0/24 [1/0] via 11.0.0.2
S    192.168.5.0/24 [1/0] via 11.0.0.2
S    192.168.6.0/24 [1/0] via 11.0.0.2
Router3#
```

- show running-config



```
!
hostname Router3
!
!
!
!
!
ip dhcp pool VOICE
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 option 150 ip 192.168.1.1
ip dhcp pool Phone
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 option 150 ip 192.168.1.1
ip dhcp pool Phone1
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 option 150 ip 192.168.1.1
ip dhcp pool Phone2
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 option 150 ip 192.168.2.1
!
!
!
```

7.3 Server/ISP Segment, DSL Modem, and External Link Simulation

An external connectivity segment was included to represent ISP access. The DSL modem connects to the switch in that segment. End devices there were configured to verify reachability.

- Devices were assigned addresses in **192.168.5.0/24**
- DSL modem was connected and configured from: DSL Modem → Config





7.4 Wireless Deployment (Public Wi-Fi + Gaming Zone)

Two access points were deployed:

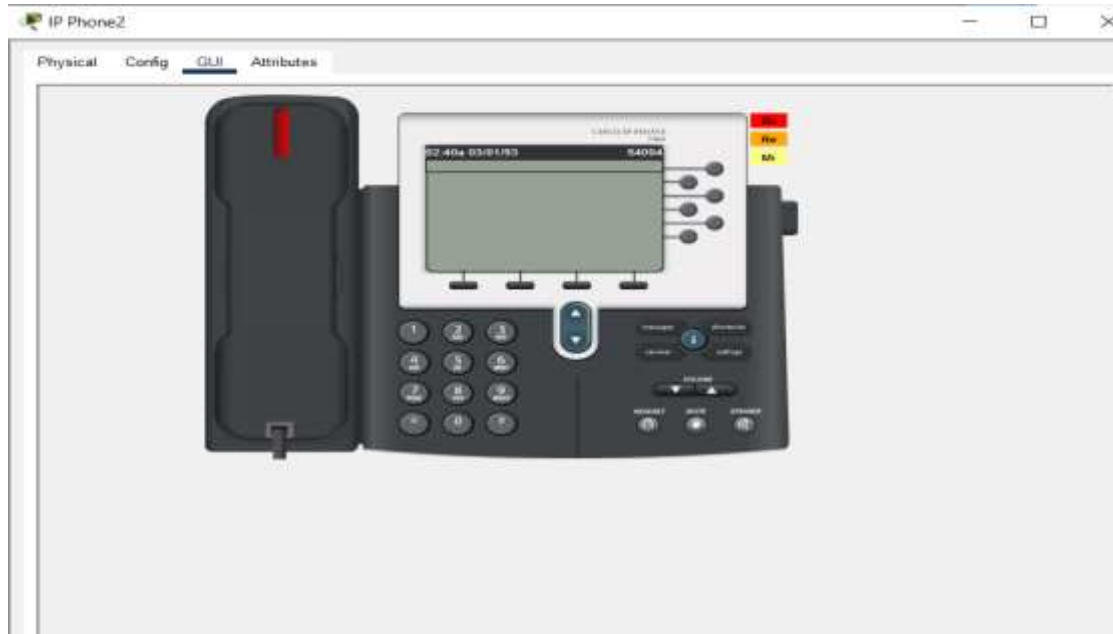
- AccessPoint1 in the Café/Public Wi-Fi area
- AccessPoint2 in the Gaming Zone

Each AP was configured with SSID and security (password), then wireless laptops were connected and tested via ping to the default gateway and remote subnets.



7.5 VoIP Enablement and Validation

IP phones were placed in multiple zones to simulate voice communication between departments/areas. After confirming IP addressing and routing, calls between phones were tested to ensure that voice endpoints were reachable.



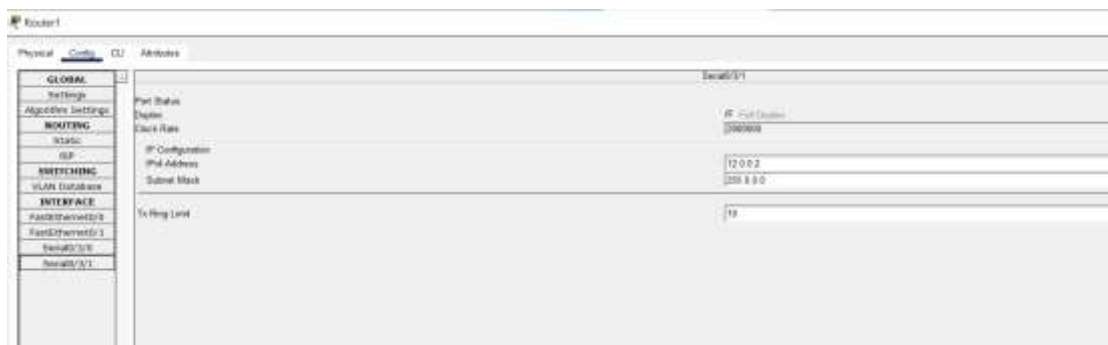


7.6 WAN Links Configuration (/30 Serial Networks)

The WAN layer used /30 networks to connect routers:

- 11.0.0.0/30 for Router1–Router2
- 12.0.0.0/30 for Router2–Router3

Serial interfaces were assigned IP addresses and brought up, then reachability between router serial endpoints was tested.



```

!
interface Serial0/3/0
 ip address 11.0.0.2 255.0.0.0
!
interface Serial0/3/1
 ip address 12.0.0.1 255.0.0.0
 clock rate 2000000
!
interface Vlan1
 no ip address
 shutdown
!
speed auto
!
interface Serial0/3/0
 ip address 11.0.0.1 255.0.0.0
 clock rate 2000000
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
!
interface Vlan1
 no ip address

```



7.7 Network Verification and Testing

Testing included:

- Ping within the same zone (LAN local tests)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time=74ms TTL=125
Reply from 192.168.2.10: bytes=32 time=7ms TTL=125
Reply from 192.168.2.10: bytes=32 time=20ms TTL=125

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 74ms, Average = 33ms
```

```
C:\>PING 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time=2ms TTL=125
Reply from 192.168.1.6: bytes=32 time=14ms TTL=125
Reply from 192.168.1.6: bytes=32 time=14ms TTL=125
Reply from 192.168.1.6: bytes=32 time=11ms TTL=125

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 14ms, Average = 10ms
```

```
C:\>PING 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=13ms TTL=126
Reply from 192.168.3.2: bytes=32 time=11ms TTL=126
Reply from 192.168.3.2: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 14ms, Average = 9ms
```




- Ping across zones (inter-LAN tests)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Laptop8	Laptop10	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Laptop8	Laptop0	ICMP		0.000	N	2	(edit)	(delete)
	Successful	Laptop8	PC5	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Laptop...	Server0	ICMP		0.000	N	4	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC5	Laptop10	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC5	Printer2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC5	PC4	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC5	Laptop3	ICMP		0.000	N	3	(edit)	(delete)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	Server0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC6	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	Laptop10	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	Laptop10	ICMP		0.000	N	3	(edit)	(delete)

- Ping to each default gateway

```
Pinging 11.0.0.2 with 32 bytes of data:

Reply from 11.0.0.2: bytes=32 time=1ms TTL=254
Reply from 11.0.0.2: bytes=32 time=12ms TTL=254
Reply from 11.0.0.2: bytes=32 time=1ms TTL=254
Reply from 11.0.0.2: bytes=32 time=1ms TTL=254
```

```
Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=253
Reply from 192.168.1.1: bytes=32 time=3ms TTL=253
Reply from 192.168.1.1: bytes=32 time=10ms TTL=253
Reply from 192.168.1.1: bytes=32 time=2ms TTL=253
```

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=3ms TTL=253
Reply from 192.168.2.1: bytes=32 time=4ms TTL=253
Reply from 192.168.2.1: bytes=32 time=3ms TTL=253
Reply from 192.168.2.1: bytes=32 time=10ms TTL=253

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 10ms, Average = 5ms
```



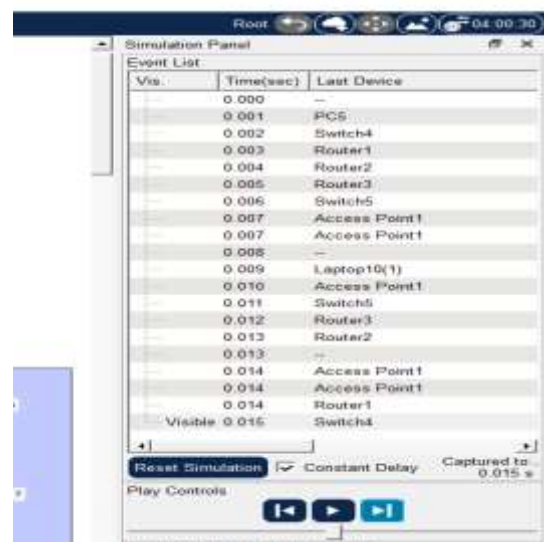
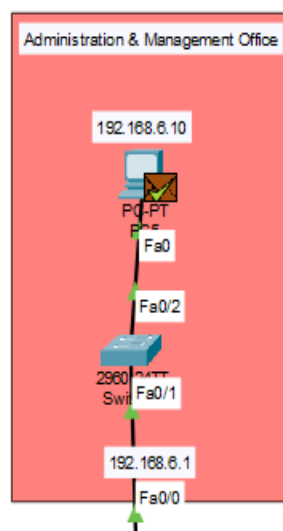

- Wireless connectivity checks (Wi-Fi clients to gateway and to remote networks)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
0	Successful	Lapto...	Router3	ICMP		0.000	N	0	(edit)	(delete)
0	Successful	Laptop0	Router2	ICMP		0.000	N	1	(edit)	(delete)
0	Successful	Lapto...	Laptop10(1)	ICMP		0.000	N	2	(edit)	(delete)

- VoIP call checks between different zones



- Simulation mode for packet flow confirmation





8. Conclusion

This project delivered a complete smart city enterprise network in Packet Tracer based on multiple star-topology LANs connected through three routers. Each functional zone was assigned its own /24 subnet, which improved organization and simplified troubleshooting. Router-to-router connectivity was achieved using two /30 serial WAN links, and routing enabled full communication between all city zones.

The final network supports wired users, public wireless connectivity, shared services such as printers and a server, and VoIP endpoints. Verification tests confirmed correct addressing, routing, wireless access operation, and overall reachability across the entire topology. The design is suitable for future expansion to include additional IoT-based traffic control devices and more smart city services.